Partial Fractions and Improper Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Partial fraction decomposition: distinct linear factors**

Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions.

$\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$. $5x-1 = A(x+2) + B(x-1)$. At $x=1$: $4 = 3A \Rightarrow A = \frac{4}{3}$. At $x=-2$: $-11 = -3B \Rightarrow B = \frac{11}{3}$. So $\frac{4/3}{x-1} + \frac{11/3}{x+2}$.

2 **[Foundational]** • **Target: Partial fraction decomposition: distinct linear factors**

Evaluate $\int \frac{3x+5}{x^2+x-2}\,dx$ using partial fractions (note $x^2+x-2 = (x+2)(x-1)$).

$\frac{3x+5}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1}$. $3x+5 = A(x-1) + B(x+2)$. At $x=1$: $8 = 3B \Rightarrow B = \frac{8}{3}$. At $x=-2$: $-1 = -3A \Rightarrow A = \frac{1}{3}$. $\int \left(\frac{1/3}{x+2} + \frac{8/3}{x-1}\right)dx = \frac{1}{3}\ln|x+2| + \frac{8}{3}\ln|x-1| + C$.

3 **[Foundational]** • **Target: Partial fraction decomposition: distinct linear factors**

Decompose $\frac{x+7}{x^2-x-6}$ into partial fractions (note $x^2-x-6 = (x-3)(x+2)$).

$\frac{x+7}{(x-3)(x+2)} = \frac{A}{x-3} + \frac{B}{x+2}$. $x+7 = A(x+2) + B(x-3)$. At $x=3$: $10 = 5A \Rightarrow A = 2$. At $x=-2$: $5 = -5B \Rightarrow B = -1$. So $\frac{2}{x-3} - \frac{1}{x+2}$.

4 **[Foundational]** • **Target: Partial fractions with a repeated linear factor**

Evaluate $\int \frac{2x+3}{(x-1)^2}\,dx$ using the decomposition found above.

$\int \left(\frac{2}{x-1} + \frac{5}{(x-1)^2}\right)dx = 2\ln|x-1| - \frac{5}{x-1} + C$.

5 **[Foundational]** • **Target: Partial fractions with an irreducible quadratic factor**

Set up (but do not solve for constants) the partial fraction decomposition of $\frac{3x^2+2}{(x-1)(x^2+4)}$.

Since x^2+4 is irreducible over the reals, the decomposition has the form $\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$.

6 **[Foundational]** • **Target: Improper integrals: infinite limits**

Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$, or show it diverges.

$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty}\left[-\frac{1}{x}\right]_1^b = \lim_{b\to\infty}\left(-\frac{1}{b}+1\right) = 0+1 = 1$. Converges to 1.

7 **[Foundational]** • **Target: Improper integrals: discontinuous integrands**

Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$, or show it diverges (note the integrand is undefined at $x=0$).

$\int_0^1 x^{-1/2}\,dx = \lim_{a\to 0^+}\left[2\sqrt{x}\right]_a^1 = \lim_{a\to 0^+}(2-2\sqrt{a}) = 2-0 = 2$. Converges to 2.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^2 \frac{1}{(x-2)^2}\,dx$, or show it diverges (the integrand is undefined at $x = 2$).

$\int_0^2 (x-2)^{-2}\,dx = \lim_{b\to 2^-}\left[-\frac{1}{x-2}\right]_0^b = \lim_{b\to 2^-}\left(-\frac{1}{b-2} - \frac{1}{2}\right) = \infty$. Diverges.

9 [Intermediate] • Target: The p-test for improper integrals

State the p-test for $\int_1^\infty \frac{1}{x^p}\,dx$, and use it to determine convergence for $p = 3$ and $p = 0.5$.

$\int_1^\infty \frac{1}{x^p}\,dx$ converges if $p > 1$ and diverges if $p \le 1$. For $p = 3 > 1$: converges. For $p = 0.5 \le 1$: diverges.

10 [Intermediate] • Target: The p-test for improper integrals

State the analogous p-test for $\int_0^1 \frac{1}{x^p}\,dx$ (a discontinuity at $x = 0$), and determine convergence for $p = \frac{1}{2}$ and $p = 2$.

$\int_0^1 \frac{1}{x^p}\,dx$ converges if $p < 1$ and diverges if $p \ge 1$ (opposite condition from the infinite-limit case). For $p = \frac{1}{2} < 1$: converges. For $p = 2 \ge 1$: diverges.

11 [Intermediate] • Target: Combining partial fractions with improper integrals

Evaluate $\int_0^\infty \frac{1}{(x+1)(x+2)}\,dx$ using partial fractions.

$\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$. $\int_0^b \left(\frac{1}{x+1} - \frac{1}{x+2}\right)dx = \left[\ln\frac{x+1}{x+2}\right]_0^b = \ln\frac{b+1}{b+2} - \ln\frac{1}{2}$. As $b \to \infty$, $\frac{b+1}{b+2} \to 1$, so the limit is $\ln 1 - \ln\frac{1}{2} = \ln 2$. Converges to $\ln 2$.

12 [Intermediate] • Target: Three distinct linear factors

Evaluate $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)}\,dx$ using the decomposition above.

$\int \left(\frac{1}{x} + \frac{1}{x-1} + \frac{4}{x+1}\right)dx = \ln|x| + \ln|x-1| + 4\ln|x+1| + C$.

13 [Intermediate] • Target: Visualizing convergence: area under the curve

The convergent improper integral $\int_1^\infty \frac{1}{x^2}\,dx = 1$ (shown earlier) means the infinite region under $y = \frac{1}{x^2}$ for $x \ge 1$ has FINITE area. Explain intuitively why this is possible even though the region extends forever.

Even though the region is infinitely long, the height $\frac{1}{x^2}$ shrinks fast enough as $x \to \infty$ that the accumulated area still converges to a finite total — the 'tail' contributes less and less area, summing to a bounded amount, similar to how a convergent geometric series sums to a finite value despite infinitely many terms.

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14 **[Intermediate]** • **Target: Visualizing convergence: area under the curve**

By contrast, $\int_1^\infty \frac{1}{x}\,dx$ diverges. Compare the graphs of $y = \frac{1}{x}$ and $y = \frac{1}{x^2}$ on $[1, 10]$ and explain why $\frac{1}{x}$ decays too slowly for the area to stay finite.

Both curves decrease, but $\frac{1}{x}$ stays larger than $\frac{1}{x^2}$ for all $x > 1$ (e.g. at $x = 10$: 0.1 vs 0.01). This slower decay means the accumulated area under $\frac{1}{x}$ keeps growing without bound as $x \to \infty$, while $\frac{1}{x^2}$'s faster decay allows its total area to converge.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Improper integrals with both an infinite bound and a discontinuity** • **Cross-Lesson: 4.02 Area in Polar Coordinates**

Evaluate $\int_0^\infty \frac{1}{\sqrt{x}(1+x)}\,dx$ is a classic doubly-improper integral (discontinuous at $x = 0$ AND infinite upper bound). Explain how such an integral must be split to evaluate it.

Split at any convenient interior point, e.g. $x = 1$: $\int_0^\infty = \int_0^1 + \int_1^\infty$. Each piece is evaluated as its own limit (one with $a \to 0^+$, the other with $b \to \infty$), and the original integral converges only if BOTH pieces converge.

16 **[Challenging]** • **Target: Comparison test for improper integrals** • **Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests**

Use a direct comparison to argue that $\int_1^\infty \frac{1}{x^2+1}\,dx$ converges, without computing it exactly.

For $x \geq 1$: $\frac{1}{x^2+1} < \frac{1}{x^2}$. Since $\int_1^\infty \frac{1}{x^2}\,dx$ converges (shown earlier, $p = 2 > 1$), and the given integrand is smaller and positive, $\int_1^\infty \frac{1}{x^2+1}\,dx$ also converges by comparison.

17 **[Challenging]** • **Target: Comparison test for improper integrals** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

Use a direct comparison to argue that $\int_1^\infty \frac{2+\sin x}{x}\,dx$ diverges.

Since $-1 \leq \sin x \leq 1$, we have $2 + \sin x \geq 1$, so $\frac{2+\sin x}{x} \geq \frac{1}{x}$ for $x \geq 1$. Since $\int_1^\infty \frac{1}{x}\,dx$ diverges, and the given integrand is at least as large, $\int_1^\infty \frac{2+\sin x}{x}\,dx$ also diverges by comparison.

18 **[Challenging]** • **Target: Combining partial fractions with improper integrals** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

Level 3 · Advanced / AP FRQThe alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ converges to sum S. If the sum is approximated by the fourth partial sum S_4, use the Alternating Series Error Bound to find an upper bound for the error $|S - S_4|$.

The series satisfies the hypotheses of the AST: $a_n = \frac{1}{n^4}$ is positive, strictly decreasing, and $\lim_{n\to\infty} \frac{1}{n^4} = 0$. By the Alternating Series Error Bound: $|S - S_4| \leq a_5 = \frac{1}{5^4} = \frac{1}{625} = \mathbf{0.0016}$. The approximation error is guaranteed to be less than or equal to 0.0016.

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19 [Challenging] • Target: Three distinct linear factors • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Level 2 · MediumApply the Ratio Test to determine the convergence or divergence of $\sum_{n=1}^{\infty} \frac{4^n}{n!}$.

Set up the ratio: $\rho = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{4^{n+1}}{(n+1)!} \cdot \frac{n!}{4^n}\right| = \lim_{n\to\infty} \left(\frac{4^{n+1}}{4^n} \cdot \frac{n!}{(n+1)n!}\right) = \lim_{n\to\infty} \frac{4}{n+1} = 0.$ Since \(\rho = 0 Ratio Test, the series converges absolutely.

20 [Challenging] • Target: Visualizing convergence: area under the curve • Cross-Lesson: 1.01 Integration by Parts

Level 1 · FoundationsExpress the repeating decimal $0.272727\ldots = 0.\overline{27}$ as an infinite geometric series, and find its exact fraction representation in simplest form.

Write as a series: $0.\overline{27} = \frac{27}{100} + \frac{27}{10000} + \frac{27}{1000000} + \ldots$ First term $a = \frac{27}{100}$, common ratio $r = \frac{1}{100}$. Since \(|r| = 0.01

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

Use a direct comparison to determine whether $\int_1^{\infty} \frac{4+\sin x}{x}\,dx$ converges or diverges.

a. Diverges only because $a = 4$ happens to be even.

b. **Diverges, since $\frac{4+\sin x}{x} \geq \frac{3}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{3}{x}\,dx$ diverges since $p = 1 \leq 1$.** ✓

c. Converges, since $\frac{4+\sin x}{x} \leq \frac{5}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{5}{x}\,dx$ converges since $p = 1 > 1$.

d. Cannot be determined by comparison, since $\sin x$ oscillates.

Since $-1 \leq \sin x \leq 1$: Diverges, since $\frac{4+\sin x}{x} \geq \frac{3}{x}$ for $x \geq 1$, and $\int_1^{\infty} \frac{3}{x}\,dx$ diverges since $p = 1 \leq 1$.

22 [Foundational] • Target: Partial fractions with a repeated linear factor

Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ using partial fractions (telescoping).

a. 2

b. diverges

c. **1** ✓

d. $\frac{1}{2}$

$\frac{1}{n(n+1)} = \frac{1}{1}\left(\frac{1}{n} - \frac{1}{n+1}\right)$. Telescoping leaves $\frac{1}{1}\left(1 + \frac{1}{2} + \cdots + \frac{1}{1}\right) = 1$.

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23 [Foundational] • Target: Partial fractions with an irreducible quadratic factor

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+6}$ converges.

a. Diverges by the nth – term test

b. Converges (compare to $\sum 1/n^2$) ✓

c. Diverges (compare to $\sum 1/n$)

d. Inconclusive

Since 01), the given series converges by the comparison test.

24 [Foundational] • Target: Improper integrals: infinite limits

Consider the infinite series $\sum_{n=1}^{\infty} \frac{n^p}{5^n}$. The series converges if and only if the parameter p satisfies which of the following conditions? Use the Ratio Test to determine the interval of convergence.

a. $p > 0$

b. $p < 1$

c. $p > -1$

d. **For all real values of** p ✓

We apply the Ratio Test. Let $a_n = \frac{n^p}{5^n}$. Then $a_{n+1} = \frac{(n+1)^p}{5^{n+1}}$. We compute the limit of the ratio of consecutive terms: $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{(n+1)^p}{5^{n+1}} \cdot \frac{5^n}{n^p}\right| \; L = \lim_{n\to\infty} \left|\frac{1}{5}\left(\frac{n+1}{n}\right)^p\right| = \lim_{n\to\infty} \left|\frac{1}{5}\left(1+\frac{1}{n}\right)^p\right|$ As $n \to \infty$.

25 [Foundational] • Target: Improper integrals: discontinuous integrands

For $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$, apply the ratio test and determine convergence.

a. Inconclusive ($L = 1$)

b. Converges ($L = 1/2 < 1$) ✓

c. Diverges ($L = 2 > 1$)

d. Diverges by comparison

$\frac{a_{n+1}}{a_n} = \left(\frac{n+1}{n}\right)^2 \cdot \frac{1}{2} \to \frac{1}{2} < 1$. Ratio test: converges regardless of the fixed power 2.

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26 [Foundational] • Target: The n-test for improper integrals

Apply the ratio test to $\sum_{n=1}^{\infty} \frac{6^n}{n!}$. What is $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$?

a. **0** ✓

b. ∞

c. 1

d. 6

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{6}{n+1} \to 0$ as $n \to \infty$. Since $L = 0 < 1$, the series converges (for every x).

27 [Foundational] • Target: Combining partial fractions with improper integrals

Determine whether $\sum_{n=1}^{\infty} \frac{n!}{n^n}$ converges using the ratio test. (variant 1)

a. Converges ($L = 1/e < 1$) ✓

b. Diverges by the nth – term test

c. Diverges ($L = 1/e > 1$)

d. Inconclusive ($L = 1$)

$\frac{a_{n+1}}{a_n} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n \to \frac{1}{e} < 1$. Ratio test: converges.

28 [Foundational] • Target: Three distinct linear factors

In the partial fraction decomposition $\frac{1}{(x-3)(x-5)} = \frac{A}{x-3} + \frac{B}{x-5}$, find A.

a. 1

b. $\frac{1}{2}$ ✓

c. $\frac{1}{3}$

d. $-\frac{1}{2}$

Multiply by $(x-3)(x-5)$: $1 = A(x-5) + B(x-3)$. Set $x = 3$: $1 = A(3-5) \Rightarrow A = \frac{1}{3-5} = \frac{1}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Visualizing convergence: area under the curve

Approximating $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ by its first 5 terms, bound the error using the alternating series error bound.

a. $|S - S_5| \leq \frac{1}{49}$

b. $|S - S_5| \leq 36$

c. $|S - S_5| \leq \frac{1}{25}$

d. $|S - S_5| \leq \frac{1}{36}$ ✓

The alternating series error bound: $|S - S_N| \leq a_{N+1} = \frac{1}{(5+1)^2} = \frac{1}{36}$.

30 [Intermediate] • Target: Improper integrals with both an infinite bound and a discontinuity

Ratio test: if $\lim \frac{a_{n+1}}{a_n} < 1$, the series:

a. Diverges

b. **Converges absolutely** ✓

c. May converge

d. Converges conditionally

Ratio < 1 implies absolute convergence.

31 [Intermediate] • Target: Comparison test for improper integrals

Repeated linear factor in partial fractions: $\frac{1}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$. Decompose $\frac{x}{(x-1)^2}$:

a. $\frac{1}{x-1} + \frac{1}{(x-1)^2}$ ✓

b. $\frac{x}{(x-1)^2}$

c. $\frac{1}{x-1}$

d. $\frac{2}{(x-1)^2}$

Applying Techniques of Integration principles, $x = (x-1) + 1 \Rightarrow A = 1, B = 1$. Therefore, the verified correct option is $\frac{1}{x-1} + \frac{1}{(x-1)^2}$.

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32 [Intermediate] • Target: Partial fraction decomposition: distinct linear factors

Consider the sequence defined by $a_n = \int_0^{1/n} \frac{\sin(x^2)}{x} dx$. Which of the following statements about the convergence of the sequence $\{a_n\}_{n=1}^{\infty}$ is true?

a. The sequence converges to 0. ✓

b. The sequence diverges to infinity.

c. The sequence converges to 1.

d. The sequence converges to $\pi/2$.

We need to determine the limit of the sequence $a_n = \int_0^{1/n} \frac{\sin(x^2)}{x} dx$ as $n \to \infty$. As $n \to \infty$, the upper limit of integration $1/n \to 0$. We examine the behavior of the integrand $\frac{\sin(x^2)}{x}$ near $x = 0$. Using the Taylor series expansion for $\sin(u)$ around $u = 0$, which is $\sin(u) = u - \frac{u^3}{3!} + O(u^5)$, we substitute $u = x^2$: $\sin(x^2) = x^2 - \frac{(x^2)^3}{3!} + O(x^{10}) = x^2 - \frac{x^6}{6} + O(x^{10})$. Therefore, for x close.

33 [Intermediate] • Target: Partial fractions with a repeated linear factor

Consider the infinite series $\sum_{n=1}^{\infty} a_n$. If the terms of the series are given by $a_n = \frac{3n+1}{2n+5}$, which of the following statements about the convergence of the series is true based on the limit of the terms?

a. The series converges because $\lim_{n\to\infty} a_n = 0$.

b. The series diverges because $\lim_{n\to\infty} a_n = \frac{3}{2}$. ✓

c. The series converges because $\lim_{n\to\infty} a_n = \frac{3}{2}$.

d. The series diverges because $\lim_{n\to\infty} a_n$ does not exist.

To determine the convergence or divergence of an infinite series $\sum_{n=1}^{\infty} a_n$, we can first examine the limit of its terms, $\lim_{n\to\infty} a_n$. If this limit is not equal to zero, then the series diverges by the Test for Divergence. For the given series, the terms are $a_n = \frac{3n+1}{2n+5}$. We evaluate the limit as $n \to \infty$: $\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{3n+1}{2n+5}$ To evaluate this limit, we can divide both the numerator and the.

34 [Intermediate] • Target: Partial fractions with an irreducible quadratic factor

Consider the infinite series $\sum_{n=1}^{\infty} \frac{n^p}{4^n}$. For what values of p does the series converge?

a. All real numbers p ✓

b. $p < 0$

c. $p > 0$

d. No real numbers p

We will use the Ratio Test to determine the convergence of the series $\sum_{n=1}^{\infty} \frac{n^p}{4^n}$. Let $a_n = \frac{n^p}{4^n}$. Then $a_{n+1} = \frac{(n+1)^p}{4^{n+1}}$. The ratio of consecutive terms is: $\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)^p/4^{n+1}}{n^p/4^n}\right| = \left|\frac{(n+1)^p}{n^p} \cdot \frac{4^n}{4^{n+1}}\right| = \left|\left(\frac{n+1}{n}\right)^p \cdot \frac{1}{4}\right| = \left|(1+\frac{1}{n})^p \cdot \frac{1}{4}\right|$ Now, we take the l.

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35 [Intermediate] • Target: Improper integrals: infinite limits

Consider the sequence $a_n = \frac{1}{n!}\int_0^n e^{-t^2}dt$. Which of the following statements about the convergence of $\{a_n\}$ is true?

a. The sequence diverges to ∞.

b. **The sequence converges to 0. ✓**

c. The sequence converges to 1.

d. The sequence oscillates and does not converge.

Let $I_n = \int_0^n e^{-t^2}dt$. We know that $e^{-t^2} \leq 1$ for all $t \geq 0$. Thus, $I_n = \int_0^n e^{-t^2}dt \leq \int_0^n 1dt = n$. More precisely, as $n \to \infty$, I_n approaches $\int_0^\infty e^{-t^2}dt = \frac{\sqrt{\pi}}{2}$, which is a finite constant. Therefore, I_n is bounded. Now consider the sequence $a_n = \frac{I_n}{n!}$. We have $0 \leq a_n = \frac{I_n}{n!} \leq \frac{n}{n!} = \frac{1}{(n-1)!}$ for $n \geq 1$. We know that $\lim_{n\to\infty} \frac{1}{(n-1)!} = 0$. By.

36 [Intermediate] • Target: Improper integrals: discontinuous integrands

Consider the polar curve given by the equation $r = 3 - 2\sin(\theta)$. A point P on this curve has polar coordinates (r, θ). If the Cartesian coordinates of P are (x, y), and $\theta = \frac{\pi}{2}$, find the value of x.

a. 3

b. **0 ✓**

c. 1

d. -1

The relationship between polar coordinates (r, θ) and Cartesian coordinates (x, y) is given by $x = r\cos(\theta)$ and $y = r\sin(\theta)$. We are given the polar equation $r = 3 - 2\sin(\theta)$ and that $\theta = \frac{\pi}{2}$. First, we find the value of r at $\theta = \frac{\pi}{2}$: $r = 3 - 2\sin(\frac{\pi}{2}) = 3 - 2(1) = 1$. Now we can find the Cartesian coordinate x using $x = r\cos(\theta)$: $x = 1 \cdot \cos(\frac{\pi}{2}) = 1 \cdot 0 = 0$. Therefore, the value of x is 0.

37 [Intermediate] • Target: The p-test for improper integrals

Consider the infinite sequence defined by $a_n = \frac{3n^2 + 2n - 1}{n^2 + 5}$. To determine if this sequence converges, we examine its limit as $n \to \infty$. Which of the following statements is true about the limit of a_n?

a. The limit is 0.

b. The limit is 2.

c. **The limit is 3. ✓**

d. The sequence diverges.

To find the limit of the sequence $a_n = \frac{3n^2 + 2n - 1}{n^2 + 5}$ as $n \to \infty$, we can divide both the numerator and the denominator by the highest power of n in the denominator, which is n^2. x As $n \to \infty$, the terms $\frac{2}{n}$, $\frac{1}{n^2}$, and $\frac{5}{n^2}$ all approach 0. $= \frac{3 + 0 - 0}{1 + 0} = \frac{3}{1} = 3$ Therefore, the sequence converges to 3.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Combining partial fractions with improper integrals** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

The polar curve is given by the equation $r = 4\cos(\theta)$. What is the equivalent Cartesian equation for this curve?

a. $x^2 + y^2 = 4x$ ✓

b. $x^2 + y^2 = 16$

c. $y = 4x$

d. $x = 4\cos(y)$

We know the relationships between polar and Cartesian coordinates: $r^2 = x^2 + y^2$, $x = r\cos(\theta)$, and $y = r\sin(\theta)$. The given polar equation is $r = 4\cos(\theta)$. To convert this to a Cartesian equation, we can multiply both sides by r to get $r^2 = 4r\cos(\theta)$. Substituting the Cartesian equivalents, we have $x^2 + y^2 = 4x$.

39 **[Challenging]** • **Target: Three distinct linear factors** • **Cross-Lesson: 6.02 Taylor and Maclaurin Series**

Consider the infinite series $\sum_{n=1}^{\infty} \frac{1}{n^p}$. For what value(s) of p does this series converge?

a. $p \leq 1$

b. $p < 1$

c. $p > 1$ ✓

d. $p = 1$

The given series is the p-series, $\sum_{n=1}^{\infty} \frac{1}{n^p}$. This series is a fundamental example in the study of convergence. The p-series test states that the p-series converges if and only if $p > 1$. Therefore, the series converges for $p > 1$.

40 **[Challenging]** • **Target: Visualizing convergence: area under the curve** • **Cross-Lesson: 1.01 Integration by Parts**

Geometric series $\sum ar^n$ converges when:

a. $r > 1$

b. $|r| < 1$ ✓

c. $r < 0$

d. $r = 1$

Applying Unit 10 · Infinite Sequences & Series — Convergence Tests principles, Convergence requires $|r| < 1$. Therefore, the verified correct option is $|r| < 1$.

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41 [Challenging] • Target: Improper integrals with both an infinite bound and a discontinuity • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Remainder estimate for alternating series: $|R_n| \le$

a. a_n

b. a_{n+1} ✓

c. $\frac{1}{n}$

d. na_n

Applying Unit 10 · Infinite Sequences & Series — Convergence Tests principles, Error ≤ first omitted term.

Therefore, the verified correct option is a_{n+1}.

42 [Challenging] • Target: Comparison test for improper integrals • Cross-Lesson: 3.01 Parametric Equations

What curve is described by the polar equation $r = 3$?

a. a line through the origin

b. a circle of radius 3 centred at the origin ✓

c. a horizontal line $y = 3$

d. a vertical line $x = 3$

$r = 3$ means every point is a constant distance 3 from the origin — a circle of radius 3.

43 [Challenging] • Target: Partial fraction decomposition: distinct linear factors • Cross-Lesson: 3.02 Vector-Valued Functions

Find the area enclosed by the polar curve $r = 14$.

a. 196π ✓

b. 14π

c. 28π

d. 98π

Area $= \frac{1}{2}\int_0^{2\pi} 14^2\, d\theta = \frac{1}{2} \cdot 196 \cdot 2\pi = 196\pi$.

44 [Challenging] • Target: Partial fractions with a repeated linear factor • Cross-Lesson: 4.01 Polar Curves and Derivatives

Interval of convergence of $\sum \frac{x^n}{n}$ is:

a. $(-1, 1)$

b. $[-1, 1)$ ✓

c. $(-1, 1]$

d. $[-1, 1]$

Radius=1; check endpoints: diverges at 1 (harmonic), converges at –1 (alternating).

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45 [Challenging] • Target: Partial fractions with an irreducible quadratic factor • Cross-Lesson: 4.02 Area in Polar Coordinates

Classify $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^3}$: absolutely convergent, conditionally convergent, or divergent?

a. Converges absolutely ✓

b. Diverges

c. Converges conditionally

d. Option 0

$\sum 1/n^3$ converges (p>1) so original converges absolutely.

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